Teacher Key: Goldfish Paradise Lab Worksheet (Basic mode)

Values below were produced by the current version of the model with the Basic settings. The model is deterministic, so students running the same settings get the same results. If the model is updated, re-check with `node tests/verification-report.mjs` or by running the lab.

Mission 1: Keep the fish alive

Trial 1 (starting tank): 0 of 8 alive. Oxygen card red (lowest about 1 mg/L). Water: pea soup. Fish start dying around day 15.

Change from Trial 1	Fish alive at day 60	Notes
Air pump On	8 of 8	Oxygen stays Healthy (lowest about 5.6 mg/L). Best single fix.
Light Low	8 of 8	Fewer algae (about 16 mg/L) means less dead algae for decomposers. Oxygen card yellow (lowest about 2.5 mg/L): a close call.
Light High	0 of 8	More algae, so a bigger crash.
Food: A little	2 of 8	Less waste, but the algae bloom still crashes the oxygen.
Food: A lot	0 of 8	Uneaten food rots, so oxygen demand rises.
6 fish / 4 fish	1 of 6 / 1 of 4	Fewer fish alone is not enough. The algae crash still happens.
2 fish	2 of 2	Very low stocking survives (lowest oxygen about 2 mg/L, right at the line).

Q2 model answer: When I turned the air pump on, the fish survived because the pump puts oxygen from the air into the water faster than the fish, algae and decomposers use it up at night.

Mission 2: Why did the fish die?

A. Days: water turns green, day 4 · decomposers multiplying, day 8 · oxygen below the danger line, day 13 · first fish died, day 15 · half the fish gone, day 19.

B. Chain order: 1 Food and fish waste add nutrients → 2 Algae grow fast → 3 Algae die and waste piles up → 4 Decomposers multiply and use up oxygen → 5 Oxygen drops very low at night → 6 Fish die.

Box numbers as printed (left to right, top row then bottom row): 4, 6, 1, 5, 2, 3.

C1: At night. Photosynthesis only happens in light, but respiration (fish, algae, bacteria) happens all the time. The daily low–high band on the chart shows this swing.

C2: Algae make oxygen only in daylight, but they use oxygen all night. When the huge population dies, decomposers use a lot of oxygen breaking the dead algae down.

C3: Mostly low oxygen (the last line of What happened? says so for the starting tank). Ammonia also climbs once fish start dying.

Mission 3: Waste to Plate

Run	Tank	Fish alive	Waste day 60 (mg N/L)	Water	Food grown (g)
1 Normal food, pump ON	Fish tank	8 of 8	about 33	pea soup	—
	+ plants	8 of 8	about 13	pea soup	about 67
2 A lot of food, pump ON	Fish tank	0 of 8	about 55	pea soup	—
	+ plants	8 of 8	about 47	pea soup	about 82
3 A lot of food, pump OFF	Fish tank	0 of 8	about 21	pea soup	—
	+ plants	0 of 8	about 1	pea soup	about 75

Watch for these: (a) The plants reduce algae (about 53 → 36 mg/L in Run 1) but do not remove it all. (b) In Run 2 the plain tank dies from toxic ammonia. With plants, ammonia stays much lower, so the fish live. (c) In Run 3 everything dies even with plants: plant leaves release oxygen into the air, not the water. Low waste numbers after a crash are misleading, because a dead tank has no fish making waste.

Matter and energy check

1 False (nitrification only changes ammonia into nitrate; the nitrogen stays). 2 True (carbon cycles). 3 False (energy flows through and leaves as heat). 4 True. 5 True.

Reflection examples

A river flows and brings in water from a whole watershed (farms, towns, rain). Temperature changes with the seasons. There are many species and sediments. Most model numbers are teaching values, not measurements.

Standards

Mission 1: HS-LS2-1 (factors affecting carrying capacity). Mission 2: HS-LS2-1, HS-LS2-4 (matter cycling and energy flow). Mission 3 and CER: HS-LS2-7 (design and evaluate a solution), HS-LS2-5 (photosynthesis and respiration in the carbon cycle). Matter and energy check: HS-LS2-4.